

CHITRANG POTDAR

Backend Engineer · Applied AI

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PROFESSIONAL SUMMARY

Backend Engineer specializing in Python backend systems and AI-enabled applications. Experience building production APIs, asynchronous workflows, multi-tenant platforms, RAG pipelines, and developer tooling with a focus on maintainable and reliable systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Flask, REST APIs, Celery

Databases & Caching: PostgreSQL, Redis, MSSQL

AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP, AWS Transcribe, Google Cloud STT, Whisper

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD

Observability: Grafana, Loki, Promtail, LangSmith

Testing & Automation: Pytest, Playwright

ML & Data: Scikit-learn, MLflow, ZenML, XGBoost

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune

Oct 2025 – Present

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Grafana · Loki · Promtail · Pytest · Playwright

- Built a multi-tenant backend with query-level data isolation across users, portfolios, and organizations — ensuring each tenant only ever accesses their own data within a shared PostgreSQL schema.
- Developed REST APIs for portfolio review, rejected order tracking, and financial calculators; integrated the Jira API to automatically create issues from application events, eliminating a manual reporting step.
- Reduced redundant database round-trips by introducing Redis-based session caching for frequently accessed user context; maintained 80%+ unit test coverage with Pytest.
- Automated NSE rejected order processing using python and end-to-end client workflows using Playwright, saving approximately 2–3 hours of manual work per week.
- Configured Promtail to ship application logs into Loki and built Grafana dashboards to surface service health and error trends — giving the team structured visibility into backend behaviour for the first time.
- Built a domain-specific MCP server to expose investment platform data to LLM clients: annotated each tool and function signature with investment-domain terminology (e.g. portfolio review, order status) so the model correctly resolves business concepts to the underlying database operations without requiring raw schema knowledge.
- Implemented async background email delivery using Redis and Celery: on task creation, a job is enqueued in Redis, a Celery worker picks it up and sends the email, and the UI receives a failure notification if delivery does not succeed.

AI Meeting Intelligence Pipeline

AWS Lambda · API Gateway · S3 · AWS Transcribe · Claude API

- Designed an async serverless pipeline to work within API Gateway's 29-second timeout: audio and agenda files are uploaded to S3, transcribed via AWS Transcribe, then passed to the Claude API to produce structured output — summary, agenda scoring, and action items.

AI Voice Conversational System with RAG

FastAPI · Google Cloud STT / Whisper · TTS · Claude API · LlamaIndex

- Built a voice AI assistant pipeline: speech-to-text → RAG retrieval (semantic chunking + embedding-based lookup) → LLM response → text-to-speech, grounding answers in a domain knowledge base.

- Designed the pipeline modularly so the knowledge base can be updated independently without changes to the core processing logic.

Python Developer Intern · ArrayPointer Pvt. Ltd., Pune

Dec 2023 – Apr 2024

- Automated structured data extraction from PDFs (electricity bills, CAS files) using PyPDF2, Camelot, and Tabula, reducing manual extraction effort by approximately 70% with JSON output.
- Built an XGBoost revenue forecasting model with incremental learning support, improving forecast accuracy by roughly 20% as new data was ingested.

PROJECTS

SOLID-Agent-Harness – Multi-Agent Code Analysis

June 2025

github.com/Dev-Chitrang/SOLID-Agent-Harness

LangChain · LangGraph · Python

- Designed a CLI-driven repository analysis tool using a LangGraph supervisor pattern: the supervisor dispatches work to four specialized agents in parallel — SOLID Auditor, Bug Hunter, Documenter, and README Generator — each independently analyzing the input codebase.
- Implemented a Report Writer node that collects outputs from all agents and compiles them into structured Markdown reports, covering SOLID principle adherence, bug analysis, architecture documentation, and a generated README — demonstrating graph-based workflow composition with LangGraph.
- Optionally used LangSmith for observability.

TaskHub – Multi-Tenant Project Management Backend

Aug – Sep 2025

github.com/Dev-Chitrang/TaskHub-Full-Stack-Project

FastAPI · PostgreSQL · Docker · GitHub Actions · Pytest

- Built a SaaS-style project management backend from scratch, implementing workspace isolation, RBAC, and a complete auth flow: email verification, TOTP-based 2FA, and password recovery.
- Containerised with Docker; deployed through a GitHub Actions CI/CD pipeline with Pytest coverage across all core API modules.

Retail DataOps & MLOps Platform

Mar – Jul 2025

github.com/Dev-Chitrang/data-warehouse-to-mlops

FastAPI · MSSQL · Scikit-learn · ZenML · MLflow · NeuralProphet

- Designed ETL pipelines using a medallion architecture (Bronze / Silver / Gold layers) on SQL Server for structured retail data processing.
- Exposed demand forecasting, customer segmentation, and recommendation models as FastAPI endpoints; managed the full ML lifecycle using MLflow for experiment tracking and ZenML for pipeline orchestration.

EDUCATION

B.E. Computer Engineering :- 8.65 CGPA

Jun 2021 – Jun 2025

Savitribai Phule Pune University, Pune

CERTIFICATIONS

- IBM AI & Cloud — AICTE Certified
- VOIS AI & Data Analysis — AICTE Certified

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis

IJIRCCE | Impact Factor: 8.65 | Nov 2024

github.com/Dev-Chitrang/Final-year-project/tree/Backend